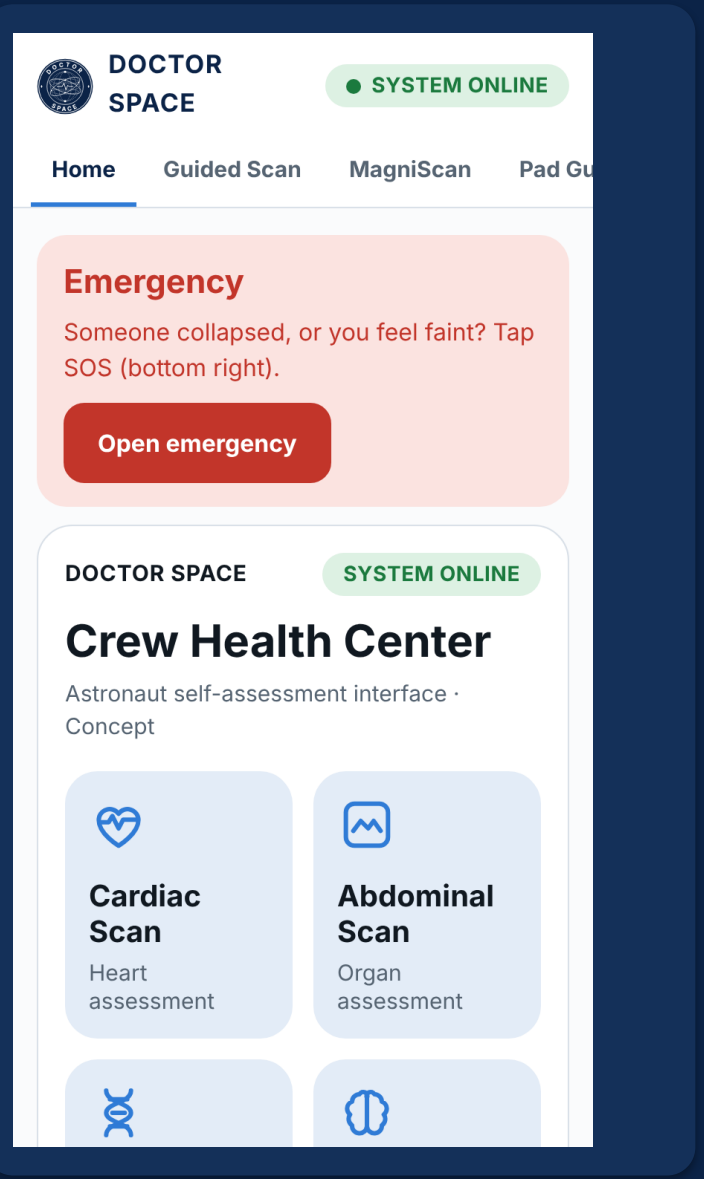
DOCTOR SPACE

Crew Health Center

A magnetic-pad ultrasound guide that lets a crew member who is not a doctor check the heart and belly, read the result, and know the next step.

Team: DOCTOR SPACE
Member: Alfathitamara Oktalia Sasti
Challenge: DOCTOR SPACE
NASA Space Apps Challenge 2026



Problem: far from Earth, no doctor

NASA lists the risks. Each one gets a tool a non-doctor can use.

NASA risk	Doctor Space tool
Cardiovascular adaptation	Cardiac scan and vitals against the astronaut’s own baseline
Spaceflight bone changes, reduced fitness	Bone Watch: magnetic heel pad and exercise log
Distance from Earth, limited medical support	Guided scan, HEALTH ALERT, SOS, locked medical kit
Space radiation	Pre-flight fertility banking planner
Inadequate sleep, behavioral changes	Sleep and mood trend in Wellness
Blood (anemia in flight)	Hemoglobin trend against baseline

The clever part is the magnet. Magnets hold the ultrasound probe on the same spot, at the same angle, for the heart, the belly and the heel.

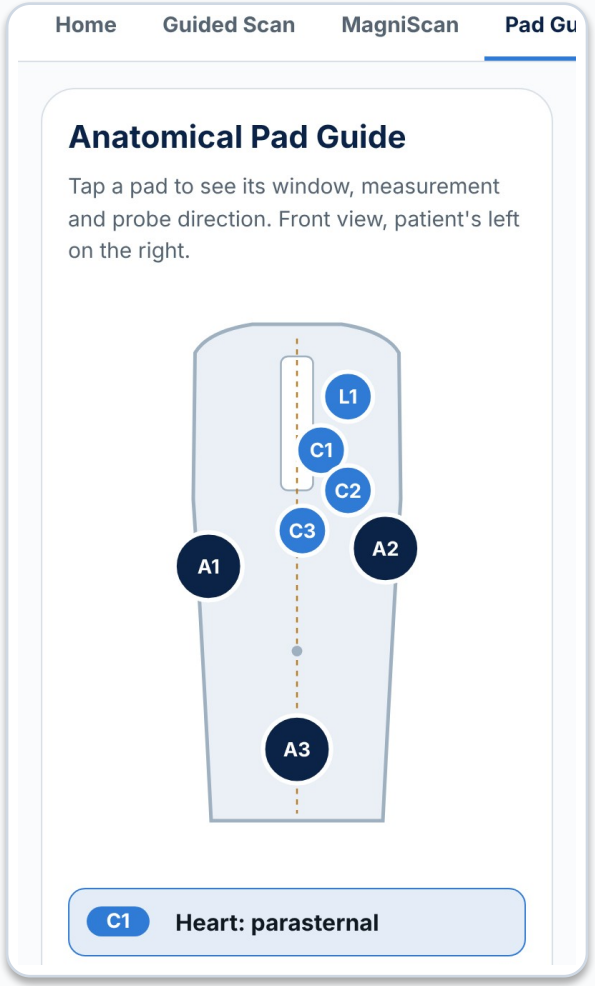
Source: NASA Human Research Program, public risk list.

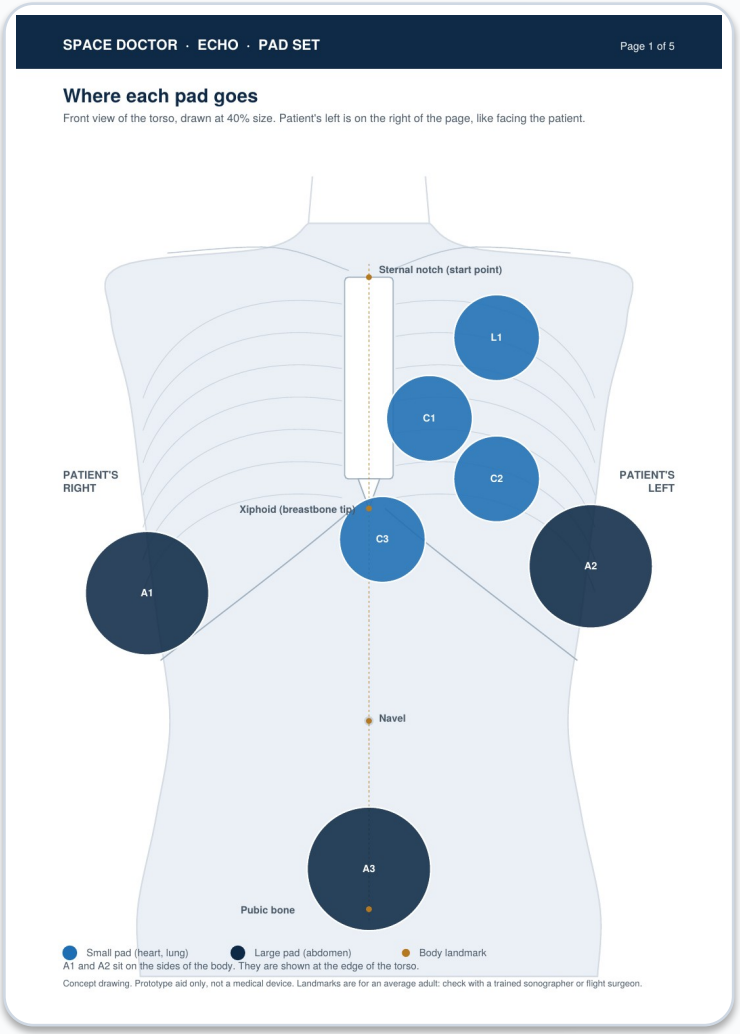
Concept prototype. Simulated data and illustrations. Not a validated medical device.

Wearable magnetic ultrasound

Sticker pads and a magnetic probe collar. Anyone can put the probe in the right place.

- **Sticker pads.** Printed pads with steel dots go on the skin. Each pad has a window for the probe.
- **Magnetic collar.** Magnets in a collar dock to the dots. They hold position and angle, not pressure.
- **Seven pads.** Heart, lung and belly, each found by measuring from the notch at the base of the throat.
- **Same spot every time.** Today’s image can be compared with the last one.
- **Low cost.** Printed sheet, steel dots, small magnets.

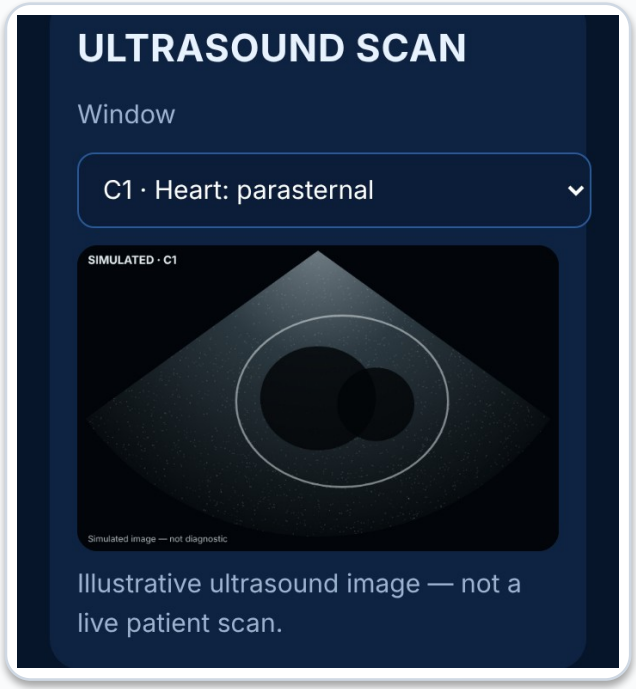




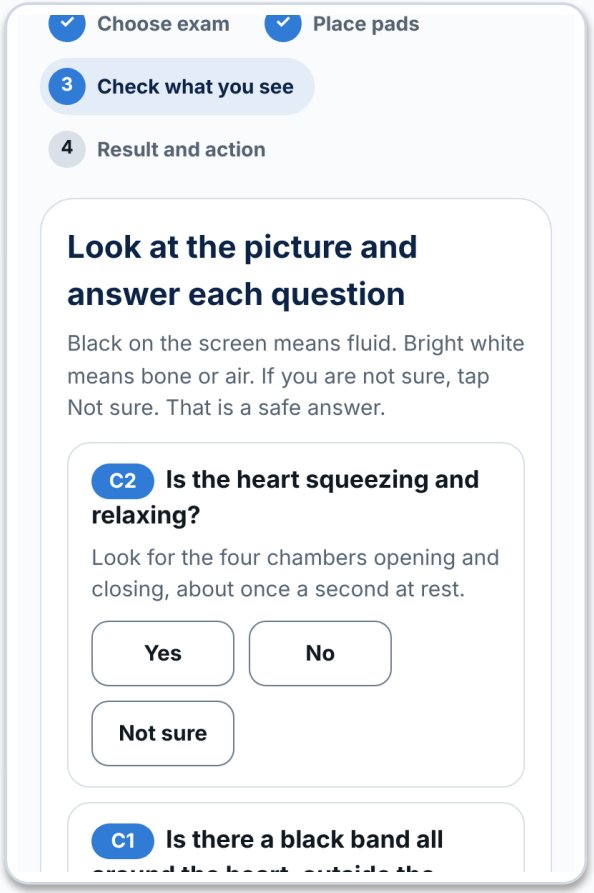
Sample positions for an average adult. A trained sonographer should confirm each window before real use.

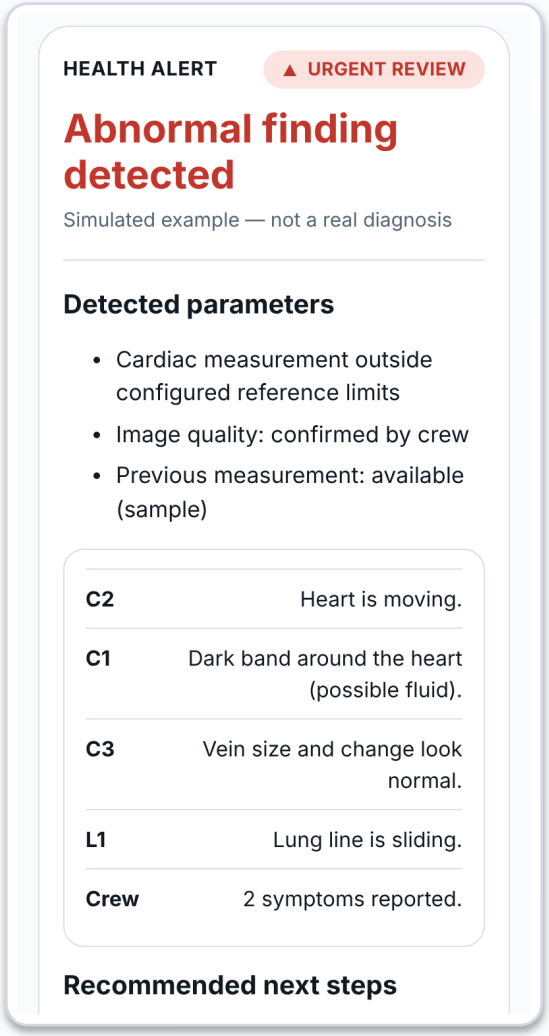
Ultrasound simulation and HEALTH ALERT

Practice the scan, answer in plain words, get one clear result.



Every image is labelled simulated. The app never picks a drug or a dose. The ground team gives the order.



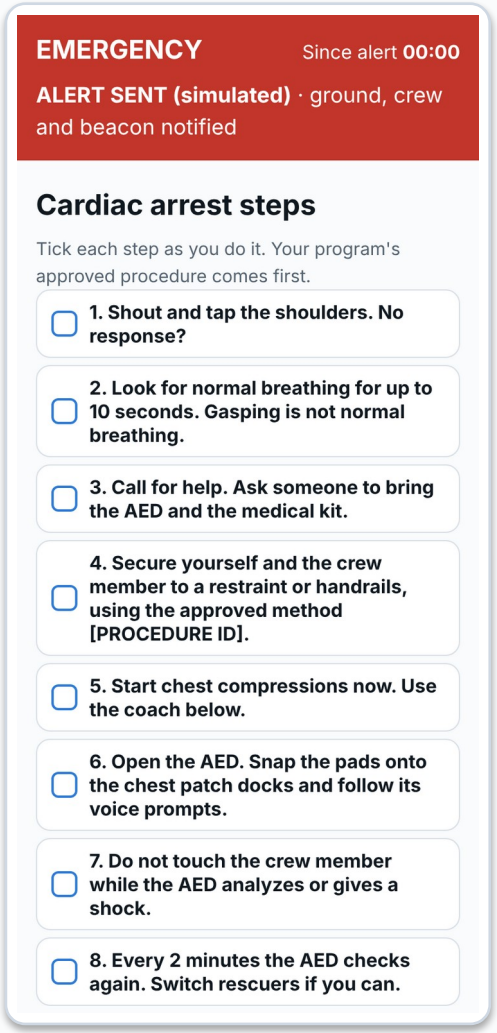


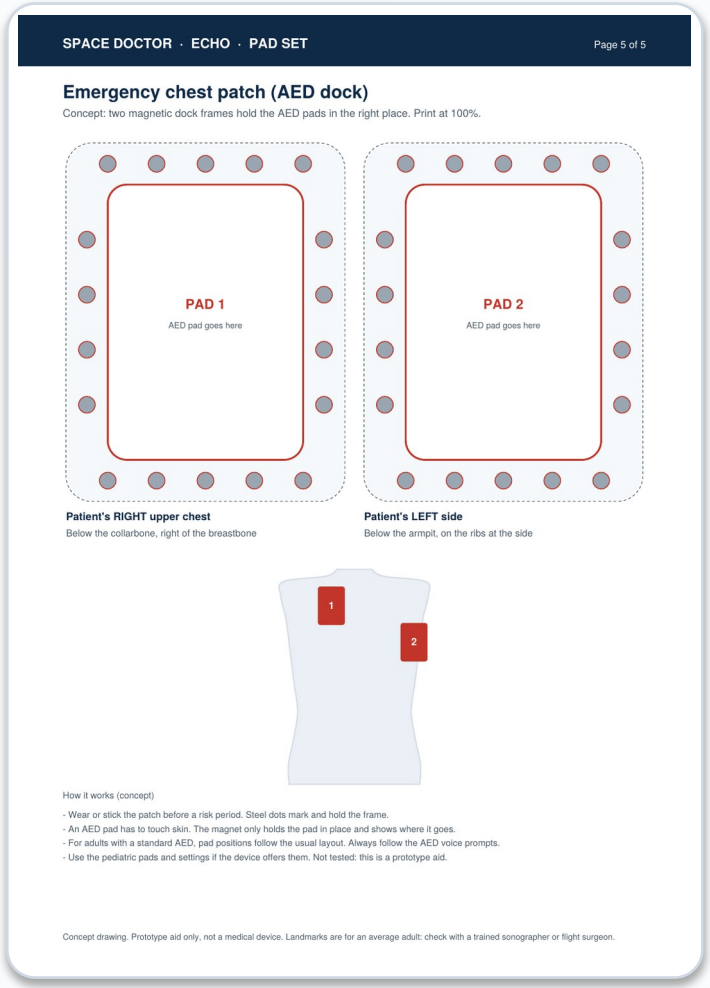
- **Normal.** All views normal.
- **Review.** One concern, or not sure.
- **Urgent.** No heart motion, or two findings.

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Guidance for the astronaut

One button for a collapse. Plain steps for someone who has never done it.





- **SOS button.** Always on screen. 10 seconds to cancel, then ground and crew are alerted.
- **CPR coach.** Checklist, then compressions: about 100 to 120 per minute, 5 to 6 cm deep, with microgravity methods.
- **AED chest patch.** Two magnetic frames hold the AED pads at the right chest positions.
- **Locked kit.** Opens only after a second person authorises it and a five-item check is ticked.
- **Why keep it.** NASA lists cardiovascular adaptation and distance from medical care as risks. SOS is the backup.

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Bone, fertility and blood

More tools for the risks on the NASA list.

Bone Watch

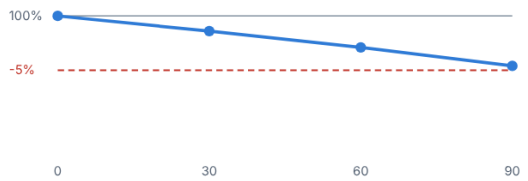
Bone changes

Without gravity, bones lose strength. NASA lists spaceflight bone change as a risk. Record a repeatable heel measurement and your exercise load. Sample data until you save your own.

● Within your limit

Latest day 90: index 95.4 (-4.6% vs baseline). Keep your routine.

Exercise last logged is below the sample target of 300 min per week.



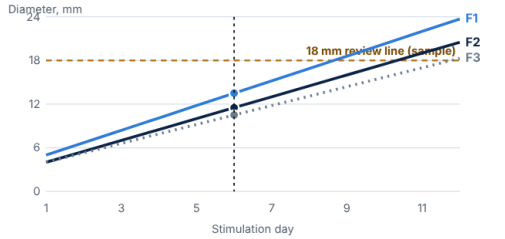
Line = heel index as a percent of the pre-flight baseline. Dashed line = your alert limit.

Pre-flight Fertility Planner

Before launch

Radiation is one of the five hazards. Banking sperm or embryos on Earth before launch keeps the option open. This is a proposal: it is not a current NASA program. Sample data.

Stimulation day **Day 6**



■ **Keep monitoring. The lead follicle is 13.5 mm and 0 follicles are at 14 mm or more.**

Sample review rule: lead follicle at least 18 mm and three follicles at least 14 mm. The fertility doctor decides timing and treatment.

- **Bone Watch.** A magnetic heel pad repeats the same measurement. The app shows the drop from the pre-flight baseline. Magnets only fix position. They do not strengthen bone.
- **Pre-flight fertility.** Banking before launch, because radiation is a hazard. Nothing is frozen in flight. A proposal, not a NASA program.
- **Blood.** Hemoglobin trend. Anemia in flight is not usually low iron, so no iron without a doctor.

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Thank you

Doctor Space gives a crew member without medical training a way to look, understand, and act.

Impact

- Space: a health backup for crews far from Earth
- Earth: remote clinics and first responders can use the same pad guide
- Free, offline, runs in any browser
- Next: test with a sonographer, then with real crew data

NASA resources used

- 5 Hazards of Human Spaceflight (NASA HRP): distance from Earth, isolation and closed environments shaped the scope
- Risks of Human Spaceflight (NASA HRP): cardiovascular adaptation, sleep and behavior set the vitals tracked
- Ongoing HRP Research in Space

Also: published ISS studies and CSA science pages

Live app: https://alfathitamara.github.io/doctor-space/

Built with help from AI tools (code, text, graphics). Illustrations and sample data are simulated. Not a validated medical device.